Year 10 Mathematics

Number 3

6.2 Negative and Fractional Indices

Negative and Fractional Indices

STARTER · 5 MIN

Date: _____ **Title:** Negative and Fractional Indices

LEARNING OBJECTIVES

- Understand and use negative indices
- Understand and use fractional indices
- Evaluate expressions with negative and fractional indices

STARTER *Answer in your book. No calculators.*

1. Work out $\sqrt{36}$

2. Work out $\sqrt[3]{27}$

3. Simplify $a^5 \div a^5$

4. Work out $\frac{1}{2^3}$

Starter — Answers

STARTER

Tick or correct in green pen

1. $\sqrt{36} = \mathbf{6}$

2. $\sqrt[3]{27} = \mathbf{3}$

3. $a^{5-5} = a^{0} = \mathbf{1}$

4. $\mathbf{\frac{1}{8}}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can understand and use negative indices	☐
SILVER	I can understand and use fractional indices	☐
GOLD	I can evaluate expressions with negative and fractional indices	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Negative and Fractional Indices

INTRODUCTION · 3 MIN

A negative index means 'one over'. A fractional index means a root.

Negative index

Take the reciprocal.

$$a^{-n} = \frac{1}{a^n}$$

Unit fraction index

Take the root.

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

Fractional index

Root first, then power.

$$a^{\frac{m}{n}} = (\sqrt[n]{a})^m$$

Example

Root, power, then reciprocal.

$$8^{-\frac{2}{3}} = \frac{1}{4}$$

Examples 1 and 2 — Negative Indices

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint $a^{-n} = \frac{1}{a^n}$

Example 1

Work out

5^{-2}

Example 2

Work out

2^{-3}

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$$5^{-2} = \frac{1}{5^2}$$

$$= \frac{1}{25}$$

Tip: negative index: one over.

Example 2

$$2^{-3} = \frac{1}{2^3}$$

$$= \frac{1}{8}$$

Tip: work out the positive power, then take the reciprocal.

Task 1

PRACTICE · 9 MIN

BRONZE *Work out the value of each, as a fraction*

Hint $a^{-n} = \frac{1}{a^n}$

1. 3^{-2}
2. 4^{-1}
3. 10^{-3}
4. 2^{-5}
5. 6^{-2}

★ Write as a power of 2: $\frac{1}{16}$

Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1 $\frac{1}{3^2} = \mathbf{\frac{1}{9}}$

2 $\mathbf{\frac{1}{4}}$

3 $\frac{1}{10^3} = \mathbf{\frac{1}{1000}}$

4 $\frac{1}{2^5} = \mathbf{\frac{1}{32}}$

5 $\frac{1}{6^2} = \mathbf{\frac{1}{36}}$

★ $\frac{1}{2^4} = \mathbf{2^{-4}}$

Examples 3 and 4 — Fractional Indices

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $a^{\frac{1}{n}} = \sqrt[n]{a}$

Example 3

Work out

$25^{\frac{1}{2}}$

Example 4

Work out

$8^{\frac{2}{3}}$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$25^{\frac{1}{2}} = \sqrt{25}$$

$$= 5$$

Tip: power one half means square root.

Example 4

$$8^{\frac{2}{3}} = (\sqrt[3]{8})^2$$

$$= 2^2$$

$$= 4$$

Tip: root first, then power.

Task 2

PRACTICE · 9 MIN

SILVER

Work out the value of each

Hint $a^{\frac{1}{n}} = \sqrt[n]{a}$

1. $49^{\frac{1}{2}}$

2. $64^{\frac{1}{3}}$

3. $16^{\frac{3}{4}}$

4. $27^{\frac{2}{3}}$

5. $100^{\frac{3}{2}}$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $\sqrt{49} = \mathbf{7}$

2 $\sqrt[3]{64} = \mathbf{4}$

3 $(\sqrt[4]{16})^3 = 2^3 = \mathbf{8}$

4 $(\sqrt[3]{27})^2 = 3^2 = \mathbf{9}$

5 $(\sqrt{100})^3 = 10^3 = \mathbf{1000}$

Examples 5 and 6 — Negative Fractional Indices

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $a^{-n} = \frac{1}{a^n}$ $\quad a^{\frac{1}{n}} = \sqrt[n]{a}$

Example 5

(2 marks)

Work out

$64^{-\frac{1}{2}}$

Example 6

(2 marks)

Work out

$27^{-\frac{2}{3}}$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$64^{-\frac{1}{2}} = \frac{1}{64^{\frac{1}{2}}}$$

$$= \mathbf{\frac{1}{8}}$$

Tip: deal with the minus sign last: one over.

Example 6

$$27^{-\frac{2}{3}} = \frac{1}{27^{\frac{2}{3}}}$$

$$= \frac{1}{3^2}$$

$$= \mathbf{\frac{1}{9}}$$

Tip: root, power, then reciprocal.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $a^{-n} = \frac{1}{a^n}$ $\quad a^{\frac{1}{n}} = \sqrt[n]{a}$

1 Work out $36^{-\frac{1}{2}}$ (2 marks)

2 Work out $8^{-\frac{1}{3}}$ (2 marks)

3 Work out $16^{-\frac{3}{2}}$ (2 marks)

4 Find n: $2^n = \frac{1}{32}$ (2 marks)

5 Find x: $9^x = 27$ (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $\frac{1}{36^{\frac{1}{2}}} = \frac{1}{6}$

2 $\frac{1}{8^{\frac{1}{3}}} = \frac{1}{2}$

3 $\frac{1}{16^{\frac{3}{2}}} = \frac{1}{4^3} = \frac{1}{64}$

4 $\frac{1}{32} = 2^{-5}, \quad n = -5$

5 $3^{2x} = 3^3, \quad 2x = 3, \quad x = \frac{3}{2}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Work out 4^{-2}

SILVER Work out $125^{\frac{2}{3}}$

GOLD Work out $81^{-\frac{1}{4}}$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE

$$\frac{1}{4^2} = \frac{\mathbf{1}}{\mathbf{16}}$$

SILVER

$$(\sqrt[3]{125})^2 = 5^2 = \mathbf{25}$$

GOLD

$$\frac{1}{81^{\frac{1}{4}}} = \frac{\mathbf{1}}{\mathbf{3}}$$

Exam Practice — Negative Indices

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $a^{-n} = \frac{1}{a^n}$

Question 1

(2 marks)

Work out the value of

(a) (1)

4^{-2}

.....

(b) (1)

10^{-3}

.....

Question 2

(2 marks)

(a) Write as a power of x (1)

$\frac{1}{x^5}$

.....

(b) Simplify (1)

$x^3 \times x^{-7}$

.....

Exam Practice — Fractional Indices

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $a^{\frac{1}{n}} = \sqrt[n]{a}$

Question 3

(2 marks)

Work out the value of

(a) (1)

$25^{\frac{1}{2}}$

.....

(b) (1)

$27^{\frac{1}{3}}$

.....

Question 4

(4 marks)

Work out the value of

(a) (2)

$8^{\frac{2}{3}}$

.....

(b) (2)

$16^{\frac{3}{4}}$

.....

Exam Practice — Negative Fractional Indices

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $a^{-n} = \frac{1}{a^n}$ $\quad a^{\frac{1}{n}} = \sqrt[n]{a}$

Question 5

(4 marks)

Work out the value of

(a) (2)

$9^{-\frac{1}{2}}$

.....

(b) (2)

$32^{-\frac{2}{5}}$

.....

Question 6

(3 marks)

Work out the exact value of

$$\left(\frac{27}{8}\right)^{-\frac{2}{3}}$$

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $\frac{1}{16}$ — B1

(b) $\frac{1}{1000}$ or 0.001 — B1

Tip: flip, then square.

Question 2

(a) x^{-5} — B1

(b) x^{-4} — B1

Tip: 3 + (−7) = −4.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) $\sqrt{25} = 5$ — B1

(b) $\sqrt[3]{27} = 3$ — B1

Tip: power a half = square root, a third = cube root.

Question 4

(a) $\sqrt[3]{8} = 2,\ 2^2 = 4$ — M1 A1

(b) $\sqrt[4]{16} = 2,\ 2^3 = 8$ — M1 A1

Tip: root first, then power.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) $\frac{1}{\sqrt{9}} = \frac{1}{3}$ — M1 A1

(b) $\frac{1}{2^2} = \frac{1}{4}$ — M1 A1

Tip: negative → flip, then root, then power.

Question 6

Flip: power becomes + $\frac{2}{3}$ — M1

$= \left(\frac{2}{3}\right)^2$ — M1

$= \frac{4}{9}$ — A1

Tip: flip the fraction to make the power positive.